Module Description: Artificial Intelligence

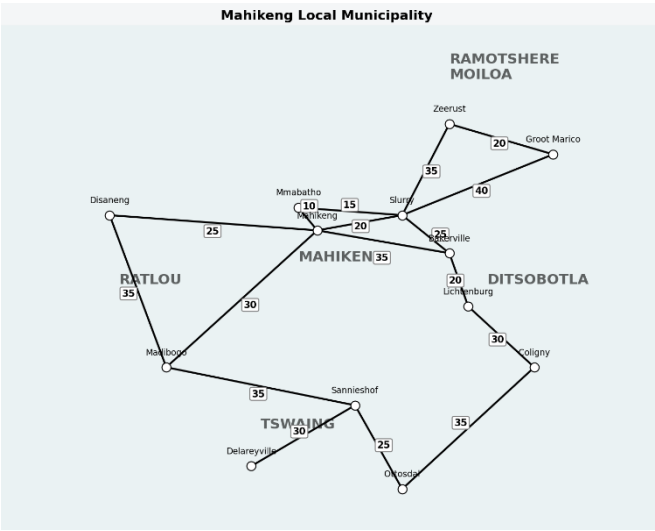
Module Code: CMPG 313

Practical Assessment: Truck Logistics Route Optimization Using A* Search

You are an **AI Engineer** working for a logistics company called **Truck Logistics**. The company wants to use **A* Search** to help delivery trucks find the **optimal route** to their destinations while reducing travel cost and improving efficiency.

In this practical, you have been given a **Python program** for A* Search **attached with this manual**. However, the code is **not complete** and contains **bugs**. Before answering the multiple-choice questions (MCQs), you must first inspect the code carefully and fix the missing and incorrect parts.

Node	Traffic per city
Disaneng	70
Mahikeng	50
Mmabatho	55
Slurry	45
Zeerust	65
Groot Marico	75
Bakerville	30
Lichtenburg	20
Coligny (Goal)	0
Ottosdal	25
Sannieshof	40
Delareyville	55
Madibogo	60



Your tasks:

1. Read the attached A* Search Python code.
2. Identify and correct the **bugs** in the program.
3. Fill in the **missing parts** of the code so that it runs correctly.
4. Execute the corrected program.
5. Use the program output to determine the **optimal route** to the destination.
6. Answer the MCQs that follow based on your corrected code and its output.
7. Submit the code and output in your github repository